Lab 5: Subtractor

Goals

- Understand arithmetic operations in the binary number system.
- Learn how to design arithmetic circuits using combinational logic.

Introduction

A binary subtractor is a combinational circuit that performs the subtraction operation between two binary numbers.

Lab description

Design an 8-bit subtractor circuit for two numbers (A and B) using two's complement with the following characteristics:

- Both inputs and outputs will have an 8-bit word width.
- The inputs will be represented in two's complement format.
- The output must be in two's complement format.
- If an overflow occurs, it must be indicated with a single bit.

Step 1

Design an adder circuit.

Step 2

Add circuitry to perform subtraction and overflow detection.

Step 3

Implement the circuit on the FPGA.

- Use switches **SW[10] to SW[17]** for input data A.
- Use switches **SW[0] to SW[7]** for input data B.
- Display input A using LEDs **LEDR[10] to LEDR[17]**.
- Display input B using LEDs **LEDR[0] to LEDR[7]**.
- Display the output data using LEDs **LEDG[0] to LEDG[7]**.
- Display the overflow bit using LED **LEDG[8]**.

Step 4

Verify the circuit operation using the following test data.

A	B	A-B (expected)	*Overflow*
36 (8'b00100100)	-127 (8'b10000001)	-93 (8'b10100011)	1
9 (8'b00001001)	99 (8'b01100011)	-90 (8'b10100110)	0
13 (8'b00001101)	-115 (8'b10001101)	-128 (8'b10000000)	1
101 (8'b01100101)	18 (8'b00010010)	83 (8'b01010011)	0
1 (8'b00000001)	13 (8'b00001101)	-12 (8'b11110100)	0

-19 (8‘b11101101)	-116 (8‘b10001100)	97 (8‘b01100001)	0
-7 (8‘b11111001)	-58 (8‘b11000110)	51 (8‘b00110011)	0
-59 (8‘b11000101)	-86 (8‘b10101010)	27 (8‘b00011011)	0
-27 (8‘b11100101)	119 (8‘b01110111)	110 (8‘b01101110)	1
18 (8‘b00010010)	-113 (8‘b10001111)	-125 (8‘b10000011)	1

Submission

Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

Video:

Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.